

Problem Set 6

Due: May 8, 2026 11:59PM ET

1 Proximal gradient and acceleration (80 points + Bonus 20 points)Consider the following ℓ_1 -regularized regression problem

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1, \quad (1)$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $\lambda > 0$.

- (10 points) Let $g(x) = \lambda|x|$ be the scaled absolute value function for $x \in \mathbb{R}$. Find the proximal operator $\text{prox}_g(x)$. Make a plot of $\text{prox}_g(x)$. Why do you think prox_g is called a soft-thresholding function?
- (10 points) Suppose $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ is proper, closed, and convex and given by $f(x) = \sum_{i=1}^n g_i(x_i)$, where $x = (x_1, \dots, x_n)$ and each $x_i \in \mathbb{R}$. Prove that the proximal operator of f is given by

$$\text{prox}_f(x) = [\text{prox}_{g_i}(x_i)]_i^n. \quad (2)$$

That is, each component of the vector $\text{prox}_f(x)$ is given by the proximal operator of g_i .

- (5 points) Apply (2) to find the proximal operator of $r(x) = \lambda \|x\|_1$ for $x \in \mathbb{R}^n$.
- (10 points) Apply the proximal gradient method to solve problem (1). Write down iteration t with a constant step size h . Please give an explicit expression of the proximal operator that you use.
- (5 points) Choose the constant step size $h = 1/L_f$, where L is the Lipschitz constant of the smooth function f in the composite optimization problem. Write down an explicit expression of L in terms of f .
- (10 points) Apply the subgradient method to solve problem (1). Write down iteration t with a constant step size h . Please give an explicit expression of the *subdifferential* of the objective function.
- (30 points) Recall gradient descent can be accelerated by Nesterov's acceleration as we discussed in Lecture 17. Proximal gradient can also be accelerated by the same principle. Suppose we have a composite optimization problem $\min_x F(x) := f(x) + r(x)$, where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L_f -smooth and convex, $r : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is proper closed and convex (may be nonsmooth), and an optimal solution exists. Then an acceleration of the proximal point method takes the following form. Initialize $x_0 = y_0$ and $\alpha_0 = 1$. In iteration t , compute

$$x_{t+1} = \text{prox}_{\frac{1}{L_f}r}(y_t - \frac{1}{L_f} \nabla f(y_t)), \quad (3)$$

$$\alpha_{t+1} = \frac{1 + \sqrt{1 + 4\alpha_t^2}}{2}, \quad (4)$$

$$y_{t+1} = x_{t+1} + \frac{\alpha_t - 1}{\alpha_t + 1}(x_{t+1} - x_t). \quad (5)$$

Generate a random matrix $A \in \mathbb{R}^{100 \times 110}$ with components independently sampled from a standard normal distribution $\mathcal{N}(0, 1)$. Let the “true” vector $x_{\text{true}} = e_3 - e_7$, where e_i is the vector with 1 in component i and 0 elsewhere. Let $b = Ax_{\text{true}}$. Set $\lambda = 1$.

For the following questions: please submit your executable codes on Assignment and the required plots on Gradescope. You can code in any language. We recommend Python. You are not allowed to use existing libraries to assist modeling or solving such as CVXPY, PyTorch, sklearn, etc.

- (a) (10 points) Implement the subgradient method with Polyak’s step size rule $x_{t+1} = x_t - h_t g_t$, where $g_t \in \partial F(x_t)$ and $h_t = \frac{F(x_t) - F^*}{\|g_t\|^2}$ if $g_t \neq 0$, otherwise $h_t = 1$. Here F^* is the optimal objective value. Since ℓ_1 -regularized linear regression promotes sparsity, we would expect the optimal solution of (1) is very close to $x_{\text{true}} = e_3 - e_7$, so you can use the objective value of x_{true} as a surrogate of F^* .
- (b) (10 points) Implement the proximal gradient method that you designed in question 4 with the L_f that you computed in question 5.
- (c) (10 points) Run 200 iterations of all the subgradient and proximal gradient algorithms on the same randomly generated A and the associated b , starting from the initial vector $x_0 = (1, 1, \dots, 1)$. Plot $F(x_t) - F^*$ over the 200 iterations of the two algorithms. Plot $\|x_t - x_{\text{true}}\|_2$ over the 200 iterations for the two algorithms. Summary your observations of the behaviors of the three algorithms.
- (d) (Bonus 10 points) Implement the accelerated proximal gradient method with the same choice of L_f as the proximal gradient method.
- (e) (Bonus 10 points) Run 200 iterations of the accelerated proximal gradient algorithms on the same randomly generated A and the associated b as the other two algorithms, starting from the initial vector $x_0 = (1, 1, \dots, 1)$. Plot $F(x_t) - F^*$ over the 200 iterations of the three algorithms. Plot $\|x_t - x_{\text{true}}\|_2$ over the 200 iterations for the three algorithms. Summary your observations of the behaviors of the three algorithms.

2 Self-concordance theory (20 points + Bonus 30 points)

Lecture results you may cite

You may freely use the following results from lecture. Throughout, f is self-concordant on its open convex domain, $\nabla^2 f \succ 0$, and $\|h\|_x := \sqrt{h^\top \nabla^2 f(x) h}$, $\|g\|_{x,*} := \sqrt{g^\top [\nabla^2 f(x)]^{-1} g}$.

Theorem (Definition / 1D form). f is self-concordant if and only if for every $x_0 \in \text{dom} f$ and every $d \in \mathbb{R}^n$, the function $\varphi(\lambda) := f(x_0 + \lambda d)$ satisfies

$$(\varphi'''(\lambda))^2 \leq 4 (\varphi''(\lambda))^3 \quad \text{for all valid } \lambda.$$

By translation invariance, it suffices to verify this at $\lambda = 0$ for every x_0 .

Theorem (Hessian stability / Dikin ellipsoid). For $r := \|y - x\|_x < 1$,

$$(1 - r)^2 \nabla^2 f(x) \preceq \nabla^2 f(y) \preceq \frac{1}{(1 - r)^2} \nabla^2 f(x).$$

Theorem (Local-norm change). For $r := \|y - x\|_x < 1$ and any h, g ,

$$\|h\|_y \leq \frac{1}{1-r} \|h\|_x, \quad \|g\|_{y,*} \leq \frac{1}{1-r} \|g\|_{x,*}.$$

Theorem (Cubic remainder bound). Let $q_x(y) := f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2} \|y - x\|_x^2$ be the second-order Taylor expansion of f at x . For $r := \|y - x\|_x < 1$,

$$|f(y) - q_x(y)| \leq \frac{r^3}{3(1-r)}.$$

Throughout, write $n(x) := -[\nabla^2 f(x)]^{-1} \nabla f(x)$ for the Newton step, and $\lambda(x) := \|\nabla f(x)\|_{x,*} = \|n(x)\|_x$ for the Newton decrement.

1. (10 points) Prove that the function $f(x) = -\log(1 - \|x\|_2^2)$ is self-concordant on the open ball $\Omega := \{x : \|x\|_2 < 1\}$.
2. (10 points) Let f be self-concordant attaining a minimum at x^* . Show that if $\|x_t - x^*\|_{x_t} < 1/2$, then a single step of Newton's method ($x_{t+1} = x_t + n(x_t)$) satisfies

$$\|x_{t+1} - x^*\|_{x_t} \leq \frac{\|x_t - x^*\|_{x_t}^2}{1 - \|x_t - x^*\|_{x_t}}.$$

Then explain why this fact alone does not imply quadratic convergence to x^* .

3. (Bonus 15 points) Let f be self-concordant. Show that if $\lambda(x_t) < 1$, then a single step of Newton's method ($x_{t+1} = x_t + n(x_t)$) guarantees

$$\lambda(x_{t+1}) \leq \left(\frac{\lambda(x_t)}{1 - \lambda(x_t)} \right)^2.$$

Conclude that if $\lambda(x_0) \leq 1/4$, then $\lambda(x_t) \leq \frac{1}{2} (2\lambda(x_0))^{2^t}$.

4. (Bonus 15 points) Let f be self-concordant and bounded below: $\inf_x f(x) > -\infty$. Show that f has a minimizer.

Hint. Pick x with $f(x) < \inf f + \delta$ for small δ , take a small step in the Newton direction, and use the cubic remainder bound to deduce $\lambda(x)$ is small. Then apply Problem 3 to argue that Newton's method from x produces a Cauchy sequence (in some fixed norm), whose limit is a minimizer. For the Cauchy step, you may use without proof that $\sum_t \lambda(x_t) < \infty$ implies $\prod_t (1 - \lambda(x_t))^{-1} < \infty$.